MIL-A-37375A(DLA-DM)
8 March 1978
SUPERSEDING
MIL-A-37375(DSA-DM)
11 June 1976

MILITARY SPECIFICATION

ALUMINUM HYDROXIDE GEL, MAGNESIUM HYDROXIDE, AND SIMETHICONE SUSPENSION

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Aluminum Hydroxide Gel, Magnesium Hydroxide and Simethicone Suspension.

1.2 Classification. The suspension covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - Regular Strength
Strength 2 - Double Strength

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Aluminum Hydroxide Gel, Magnesium Hydroxide and Simethicone Suspension and shall contain the following ingredients per each 5 ml, in addition to suitable flavoring and preservative agents:

	Strength 1	Strength 2
Aluminum Hydroxide Gel	Approx. 200 mg	Approx. 400 mg
Magnesium Hydroxide	Approx. 200 mg	Approx. 400 mg
Simethicone	Approx. 20 mg	Approx. 30 mg

The actual amounts of the above ingredients may vary, depending upon the aluminum oxide, magnesium and simethicone assays (see 3.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

3.1.1 Description. The finished suspension shall be a homogeneous oral suspension with a pleasant and palatable taste (see 3.1.7).

3.1.2 Assay.

3.1.2.1 Aluminum oxide content.

3.1.2.1.1 Strength 1. For strength 1, the aluminum oxide content in each 5 ml of finished suspension shall be not less than 90 mg and not more than 115 mg when determined as specified in 4.7.1.

3.1.2.1.2 Strength 2. For strength 2, the aluminum oxide in each 5 ml of finished suspension shall be not less than 180 mg and not more than 230 mg when determined as specified in 4.7.1.

3.1.2.2 Magnesium content.

3.1.2.2.1 Strength 1. For strength 1, the magnesium content in each 5 ml of finished suspension shall be not less than 80 mg and not more than 95 mg when determined as specified in 4.7.2.

3.1.2.2.2 Strength 2. For strength 2, the magnesium content in each 5 ml of finished suspension shall be not less than 150 mg and not more than 192 mg, when determined as specified in 4.7.2.

3.1.2.3 Simethicone content.

3.1.2.3.1 Strength 1. For strength 1, the simethicone content in each 5 ml of finished suspension shall be not less than 18 mg and not more than 25 mg when determined as specified in 4.7.3. As an alternate, the simethicone content in each 5 ml of the finished suspension shall be not less than 22 mg and nor more than 32.5 mg when tested as specified in 4.7.3.

3.1.2.3.2 Strength 2. For strength 2, the simethicone content in each 5 ml of finished suspension shall be not less than 27 mg and not more than 37.5 mg when determined as specified in 4.7.3.

3.1.3 pH. For strength 1 and strength 2, the pH of the finished suspension shall be not less than 7.50 and not more than 8.50 when determined potentiometrically at 25°C using the USP method.

3.1.4 Acid neutralizing capacity. The acid neutralizing capacity shall be in accordance with the requirement and test for Acid Neutralizing Capacity as specified in the Over-the-Counter Drugs Monograph for Antacid and Antiflatulent Products.

3.1.5 Microbial limits. For strength 1 and strength 2, microbial limits shall comply with the limits of the USP for Alumina and Magnesia Oral Suspension, when tested by the applicable USP method.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

3.1.6 Defoaming activity. For strength 1 and strength 2, the collapse time for the foam shall not exceed 45 seconds when determined as specified in 4.7.6.

3.1.7 Flavor and palatability. The suspension shall be pleasantly flavored, and shall be palatable and pleasant to the taste with no unpleasant aftertaste. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) individually packaged samples (each containing 5 fl oz) of the finished suspension, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.7.7 - Palatability test). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the palatability requirement.

3.2 Ingredients.

3.2.1 Aluminum hydroxide gel. The aluminum hydroxide gel used in the manufacture of the finished suspension shall be in accordance with the tests, standards, and requirements of the USP, including any supplements or revisions thereto. As an alternate to USP grade material, the aluminum hydroxide gel shall comply with the following:

3.2.1.1 Identification. Shall comply with the identification test in the USP monograph for aluminum hydroxide gel.

3.2.1.2 Microbial limits. Shall comply with the requirements in the USP monograph for aluminum hydroxide gel when determined using the USP method.

3.2.1.3 Acid neutralizing capacity. The acid neutralizing capacity shall be in accordance with the requirement and test for acid neutralizing capacity as specified in the Over-the-Counter Drugs Monograph for Antacid and Antiflatulent Products.

3.2.1.4 Assay. Each 100 grams shall contain the equivalent of not less than 8.5 grams and not more than 9.5 grams of aluminum oxide, in the form of aluminum hydroxide and hydrated oxide, when tested as specified in the USP monograph for aluminum hydroxide gel. The sample size and dilutions may be modified to accommodate a concentrated gel.

3.2.2 Magnesium hydroxide. The magnesium hydroxide used in the manufacture of the finished suspension shall be in accordance with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

3.2.2.1 Magnesium hydroxide paste. As an alternate to Magnesium Hydroxide, NF, a magnesium hydroxide paste may be utilized in the preparation of the finished suspension. The magnesium hydroxide paste shall be prepared using magnesium hydroxide complying with the tests, standards, and requirements of the NF. All other ingredients entering into the manufacture of the paste shall be of USP or NF quality, if monographed therein, and shall be of the highest pharmaceutical grade. The magnesium hydroxide paste shall comply with the following:

3.2.2.1.1 Description. Shall be a smooth paste containing 30 percent magnesium hydroxide within the designated assay limits.

3.2.2.1.2 Identification. Shall comply with the identification test in the NF monograph for magnesium hydroxide.

3.2.2.1.3 Assay. Shall assay to contain not less than 29.0 percent and not more than 33.0 percent magnesium hydroxide when assayed using the NF method for magnesium hydroxide. The sample size and dilutions may be modified to accommodate the paste.

3.2.3 Simethicone. The simethicone used in the manufacture of the finished suspension shall be in accordance wtih the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.2.3.1 Simethicone suspension. As an alternate to Simethicone, NF, a simethicone suspension may be utilized in the preparation of the finished suspension. The simethicone suspension shall be prepared using simethicone complying with the tests, standards, and requirements of the NF. All other ingredients entering into the manufacture of the simethicone suspension shall be of the USP or NF quality, if monographed therein, or shall be of the highest pharmaceutical grade.

3.3 Added substance. Added substances may be included to assure a suitable, stable product. Such substances shall be nontoxic and harmless in the amounts administered and shall not interfere with the therapeutic efficacy of the finished suspension or with the tests and assays specified herein. In addition, when such added ingredients are used in the manufacture of the finished suspension, they shall be of USP or NF quality or, if not included in either of these compendia, they shall be suitable for use in this suspension.

3.4 Label quantity of simethicone. For strength 1 material having a simethicone content in each 5 ml of the finished preparation of not less than 22 mg and not more than 32.5 mg, the quantity of simethicone specified on the label shall be no greater than 25 mg.

3.5 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

3.6 Workmanship. Workmanship shall be first class throughout. The material and its containers shall be free from any defects which detract from their appearance or impair their serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of suspension produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

Visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Bottle
Minor	II	2.5	Bottle

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Aluminum oxide content	3.1.2.1		5 composite
Magnesium content	3.1.2.2		5 composite
Simethicone content	3.1.2.3		5 composite
pH	3.1.3	S-2	5
Microbial limits	3.1.5		5
Defoaming activity	3.1.6		5 composite
Active ingredients:			
Aluminum hydroxide gel	3.2.1	2/	3/
Magnesium hydroxide	3.2.2		
Simethicone	3.2.3		

1/ Where possible, the same sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.
3/ No minimum sample is required.

4.4 Examination. The suspension shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Suspension not homogeneous after shaking.
102	Suspension not free from foreign matter.
103	Suspension not pourable from immediate container.
Minor	
201	Exterior of immediate container not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Aluminum oxide content. The aluminum oxide content in the finished suspension shall be determined as follows:

Reagents.

1. 0.05M Disodium ethylenediaminetetraacetate (EDTA). Standardized as per USP.

2. Ammonium acetate buffer. Dissolve 77.1 grams of ammonium acetate in purified water, add 57 ml of glacial acetic acid, and dilute with purified water to 1000 ml.

3. SDA-3A alcohol.

4. Dithizone. Dissolve 25.6 mg of dithizone in 100 ml of SDA-3A alcohol. Store in refrigerator and prepare fresh every two months.

5. 0.05M zinc sulfate. Dissolve 14.4 grams of zinc sulfate, heptahydrate ($ZnSO_4.7H_2O$) or 9.0 grams of zinc sulfate, dried ($Zn\ SO_4.H_2O$) in sufficient purified water to make 1 liter. Standardize as follows: Pipet 10 ml of standardized EDTA into a 125 ml Erlenmeyer flask and add in the order given: 10 ml ammonium acetate buffer, 25 ml of SDA-3A alcohol and 1 ml dithizone. Titrate with the zinc sulfate solution to a clear rose-pink color. Calculate the molarity of the zinc sulfate solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

Procedure:

Accurately weigh approximately 10 ml of strength 1 sample or 5 ml of strength 2 sample into a 150 ml beaker. Add 50 ml of purified water and 5 ml of concentrated HCl to the beaker and boil for 15 minutes. Filter if necessary into a 100 ml volumetric flask, cool to room temperature, dilute to the mark with purified water and shake. Pipet 10 ml of this solution into a 150 ml beaker, add a teflon stirring bar and place the beaker on a magnetic stirrer. Add, in the following order, with constant stirring, 15.0 ml of EDTA, 10 ml of ammonium acetate buffer, 25 ml of SDA-3A alcohol and 1 ml of dithizone. Titrate with 0.05M Zn SO_4 solution to a color change from blue-gree to rose-pink. Perform a blank titration substituting 10 ml of purified water for the sample.

Calculate the amount of aluminum oxide present in the sample as follows:

Calculation:

$$\frac{(V_B - V_S) \times M \times 50.98 \times \frac{100}{10} \times W_T}{0.05M \qquad\qquad\qquad\qquad\qquad\qquad W_S} = \frac{\text{mg aluminum oxide}}{5 \text{ ml}}$$

Where:

V_B = ml of 0.5M zinc sulfate required by the blank
V_S = ml of 0.05M zinc sulfate required by the sample
M = actual molarity of the zinc sulfate solution
50.98 = equivalent weight of aluminum oxide
$\frac{100}{10}$ = dilution factor
W_T = weight of 5 ml of sample (grams)
W_S = weight of sample used (grams)

4.7.2 Magnesium content.

Reagents.

1. 0.05M Disodium ethylenediaminetetraacetate, (EDTA). Standardize as per USP.

2. Triethanolamine solution. Dilute triethanolamine, reagent grade, with purified water in a 1:1 ratio.

3. Ammonia Buffer solution. Dissolve 6.75 grams of ammonium chloride, reagent grade, in 30 ml of purified water. Add 57 ml of ammonium hydroxide, 28 percent analytical reagent and dilute to 100 ml with purified water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4. Eriochrome Black-T indicator. Dissolve 500 mg of Eriochrome Black-T and 4.5 grams of hydroxylamine hydrochloride in methanol to make 100 ml of solution.

Procedure:

Accurately weigh approximately 10 ml of strength 1 or 5 ml of strength 2 sample into a 150 ml beaker. Add 5 ml of concentrated hydrochloric acid and 50 ml of purified water to the beaker and place on a hot plate for 15 minutes. Remove the beaker and filter the solution into a 100 ml volumetric flask through type D, or equivalent, fast filter paper. Wash the filter paper with small portions of hot purified water and add these washings to the filtrate. Cool to room temperature, dilute to the mark with purified water and mix well. Pipet 10 ml of this solution into a 150 ml Erlenmeyer flask and add in the following order: 50 ml of purified water, 5 ml of triethanolamine solution, 5 ml of 0.05M EDTA solution, 4 drops of Eriochrome Black-T indicator and 5 ml of ammonia buffer. Note: This order of the addition of the reagents is important to prevent the precipitation of the magnesium from the solution. Titrate with 0.05M EDTA solution to a blue end-point. The color change is red to blue. Titrate slowly near the endpoint, as the color change occurs slowly at the endpoint. Record the total volume of EDTA used (including the original 5 ml added previous to titration) and calculate the amount of magnesium present in the sample as follows:

Calculation:

$$\frac{M \times V \times 1.216 \times \frac{100}{10} \times W_T}{0.05 \quad W_S} = \frac{\text{mg magnesium}}{5 \text{ ml}}$$

Where:

M = actual molarity of EDTA solution
V = total volume of EDTA solution, in ml
1.216 = equivalent weight of magnesium, mg/ml
$\frac{100}{10}$ = dilution factor
W_T = weight of 5 ml of sample, in grams
W_S = weight of sample used, in grams

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4.7.3 Simethicone content. The simethicone content in the finished suspension shall be determined by one of the following methods:

4.7.3.1 Method I.

Reagents.

1. Carbon tetrachloride, analytical reagent grade
2. Hydrochloric acid, 6N
3. NF dimethylpolysiloxane reference standard

Dimethylpolysiloxane NF Reference Standard

Accurately weigh approximately 1.000 gram of dimethylpolysiloxane NF reference standard into a 250 ml volumetric flask. Bring to volume with carbon tetrachloride and mix. This standard may be stored at room temperature for approximately one month. Weigh to the nearest mg into a centrifuge tube a portion of the sample, thoroughly mixed, which is equivalent to about 40 mg of simethicone. Add 10.0 ml of carbon tetrachloride and 20 ml of purified water to the tube. Shake on a reciprocating shaker for 30 minutes at high speed, and then pour the contents onto a phase separating paper and collect the carbon tetrachloride filtrate in a beaker. Use a glass syringe to fill a 0.2 mm sodium chloride sample cell with carbon tetrachloride and place the cells in an IR spectrophotometer and scan from 1350 to 1150 cm^{-1} at medium speed. Obtain the net absorbance of the C-Si stretching band occurring at about 1268 cm^{-1}. (The net absorbance is obtained by subtracting the baseline absorbance at the same wavelength, from the absorbance maxima.) Pipet 10 ml of the standard stock solution into a 50 ml centrifuge tube, add 20 ml purified water and treat the solution the same as the sample.

Calculate the simethicone content as follows:

$$\frac{\text{Absorbance of sample}}{\text{Absorbance of standard}} \times \frac{\text{conc. of std. (mg/ml)}}{0.9575} \times \frac{W_T}{W_S} = \frac{\text{mg simethicone}}{5 \text{ ml}}$$

Where:

W_T = weight in grams of 5 ml of sample
W_S = weight of sample used in grams
0.9575 = factor to convert from dimethylpolysiloxane to simethicone.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4.7.3.2 Method II.

Reagents.

1. Carbon tetrachloride, analytical reagent grade
2. 2N sodium hydroxide
3. NF polydimethylsiloxane reference standard

Polydimethylsiloxane NF Reference Standard:

Accurately weigh approximately 1.000 gram of polydimethylsiloxane NF reference standard into a 250 ml volumetric flask. Bring to volume with carbon tetrachloride and mix. This standard may be stored at room temperature for approximately one month.

Procedure:

Weigh by difference, from a syringe or pipet, into a 50 ml centrifuge tube a portion of sample equivalent to 40 mg of simethicone. Pipet 10.0 ml of carbon tetrachloride into the tube. Add 10 ml purified water to the tube. Shake on reciprocating shaker for 30 minutes at high speed. Centrifuge for approximately 5 minutes or until clear separation of the layers is obtained. Remove as much as possible of the lower carbon tetrachloride layer, using a 10 ml hypodermic syringe fitted with a blunt needle and note volume. Transfer to a 50 ml centrifuge tube. Add an equal volume of 2N sodium hydroxide and shake on a reciprocating shaker for 15 minutes. Centrifuge for 5 minutes at approximately 1500 rpm or until a clear separation of layers is obtained. Remove as much as possible of the lower carbon tetrachloride layer using a 10 ml hypodermic syringe fitted with a blunt needle.

Insert a loosely fitting small plug of plain glass wool into the stem of a funnel just below the cone. Add about 4.0 gram of anhydrous sodium sulfate crystals onto the glass wool. Dispense the carbon tetrachloride solution from the syringe directly onto the sodium sulfate and collect in glass stoppered flasks.

Use a glass syringe to fill a 0.2 mm sodium chloride sample cell with the filtrate. Fill a 0.2 mm reference cell with carbon tetrachloride and place the cells in an IR spectrophotometer and scan from 1350 to 1150 cm^{-1} at medium speed. Obtain the net absorbance of the C-Si stretching band occurring at about 1268 cm^{-1}. (The net absorbance is obtained by subtracting the baseline absorbance at the same wavelength, from the absorbance maxima.) Pipet 10 ml of the standard stock solution into a 50 ml centrifuge tube, add 20 ml purified water and treat the solution the same as the sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

Calculate the simethicone content as follows:

Calculation:

$$\frac{\text{Absorbance of sample}}{\text{Absorbance of standard}} \times \frac{40.0}{0.9575} \times \frac{W_T}{W_S} = \frac{\text{mg simethicone}}{5 \text{ ml}}$$

Where: 40.0 = mg of polydimethylsiloxane in 10 ml aliquot
W_T = Weight in grams of 5 ml of sample
W_S = Weight of sample used in grams
0.9575 = Factor to convert from polydimethylsiloxane to simethicone.

4.7.4 Total bacteria count and absence of all gram negative enteric bacilli and pseudomonas. Shall be determined as follows:

Reagents.

1. Plate count agar. Prepared according to label directions. (Soybean casein digest agar or tryptone glucose extract agar may be used.)

2. Lactose broth with neutralizers. Prepare lactose broth as directed on label. Add 4 percent W/V Polysorbate 20 or 80 to neutralize the preservatives and 0.1 percent W/V sodium thiosulfate ($Na_2S_2O_3.5H_2O$) to neutralize any residual chlorine in the sample.

Procedure: (For total bacteria count)

Place 4 ml of purified water in a 16 by 125 mm test tube and sterilize by autoclaving for 15 minutes at 15 lbs pressure. Cool samples to room temperature (approximately 25°C) before beginning microbiological testing. Shake the suspension (sample) thoroughly and aseptically transfer a 1.0 ml aliquot to a 16 by 125 mm test tube containing 4 ml of sterile purified water. Mix the sample using standard microbiological procedures. Pipet 1.0 ml of the diluted sample into a sterile Petri dish and aseptically pour a sufficient amount of sterile plate count agar (cooled to approximately 45°C) into the dish. Cover and mix the sample by swirling. When the agar has hardened, invert the plate and incubate for two (2) days at 35°C ± 2°C. Remove the plate and count the number of organisms on each plate.

Procedure: (For absence of all gram negative enteric bacilli and Pseudomonas.)

Using a sterile 10 ml pipet, aseptically inoculate 10 ml of the finished suspension into a test tube containing lactose broth with neutralizers. Incubate these broth tubes under the same conditions as the agar plates. After incubation, check the broth cultures for turbidity. The tubes shall be clear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

4.7.5 Mold contents. The mold content shall be determined as follows:

Proceed as for the bacteria count (4.7.4) except use Sabouraud's dextrose or maltose agar and incubate at 25° - 30°C for 5 days.

4.7.6 Defoaming action. The defoaming action shall be determined as follows:

Reagents.

1. Triton X-100
2. FD&C Blue #1
3. Hydrochloric Acid, concentrated

Procedure:

Preparation of foaming solution.

Prepare a 1 percent (W/V) solution of Triton X-100 in purified water. Add 8.3 of concentrated hydrochloric acid and 5 mg of FD&C Blue #1 to each liter of solution.

Pour 100 ml of the foaming solution into a jar, cap, and place in a 37°C ± 2°C water bath for about 20 minutes. Remove the jar from the water bath and clamp into position on a shaker 1/. Add 5.0 ml of strength 1 sample or 3.3 ml of strength 2 sample (previously shaken well) directly to the jar, cap and shake for 10 seconds using the number 10 setting in a Burrell wrist action shaker, model BB or other equivalent shaker. Determine the foam collapse time in seconds. The collapse time is taken when the first portion of the foam fill liquid surface can be observed.

1/ The shaker clamp should be adjusted so that the distance between the center of the shaft and the center of the jar is 5-1/4 inches.

4.7.7 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded and served to panel members, under controlled serving conditions; e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied amoung panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the palatability standard 1/, when determined by the

1/ The palatability standard is available upon separate request to the contracting officer, Defense Personnel Support Center.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A, B or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing forty-eight bottles of the strength specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Forty-eight bottles of one strength only shall be packaged in a box conforming to PPP-B-636, class weather-resistant. The box shall include partitions, liner, and top and bottom pads.

5.2.2.2.1 Partitions. Partitions shall be full or shoulder height half slotted individual snug-fitting cell for each bottle. Partitions shall not be required when plastic bottles are utilized.

5.2.2.2.2 Liners. Liners shall be one piece covering the sides and ends of the box and fabricated of the same material as the box. Liners shall be of the same height as the box.

5.2.2.2.3 Pads. Top and bottom pads shall be fabricated of the same material as the box and not more than 1/8 inch less than the inside length and width of the box, and positioned on bottom and top of the liners and partitions.

5.2.2.2.4 Closure. Closure shall be as specified in the appendix of the box specification. When water resistant boxes are used, the box shall be waterproofed and strapping shall also be required.

5.2.3 Level B. Forty-eight bottles of one strength only shall be packaged as specified in 5.2.2.2 except that the box shall conform to PPP-B-636, class domestic.

5.2.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

5.3 Packing. Units packaged as specified in 5.2 shall require no further packing.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) When the unit package is also the exterior container (see 5.2.3 Level B) the exterior container markings as specified in 5.4.2 shall apply in lieu of the above specified markings.

5.4.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. The lot (control) number and the following legend shall be shown:

"DO NOT PERMIT TO FREEZE"

NOTE: When the unit package is also the exterior container (see 5.2.3), the level of protection shall be shown as "A" in lieu of "A/A"; "B" in lieu of "A/B"; "C" in lieu of "C/C".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

6.2 This specification covers the following items:

National Stock No.	Item Identification	Specification Designation
6505-00-080-0975	ALUMINUM HYDROXIDE GEL, MAGNESIUM HYDROXIDE, AND SIMETHICONE SUSPENSION, 5 fl oz (148 ml), 48s	Strength 1
6505-00-149-0247	ALUMINUM HYDROXIDE GEL, MAGNESIUM HYDROXIDE, AND SIMETHICONE SUSPENSION, Double Strength, 5 fl oz (148 ml), 48s	Strength 2

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
DLA - DM

Project No. 6505-1785

☆ U.S. GOVERNMENT PRINTING OFFICE: 1978 — 703-122/2324

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE ALUMINUM HYDROXIDE GEL, MAGNESIUM HYDROXIDE, MIL-A-37375A(DLA-DM) AND SIMETHICONE SUSPENSION

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:33Z
Check the source to verify that this is the current version before use.